Questions on the remaining value-group profile switch from Theorem 3.11 to Corollary 3.12

Where is it supplied? A central question with a test, and component questions on local shears and object membership

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Abstract

In Step (xi) of the proof of Corollary 3.12 of inter-universal Teichmüller theory, the q-pilot is compared with the possible images of the Θ-pilot produced by the multiradial algorithm of Theorem 3.11, and the inequality is drawn from the property (SHE) in the form stated in Step (xi-e). Several works, in abstractions of the passage and, in the author's earlier notes, for the statements that the proof actually cites, show that this form does not follow from what is stated about the output of the algorithm; in a conditional assembly, the remaining input is a value-group profile switch whose profile part is independent of every other input. Whether the construction of the algorithm itself supplies the step, and if so where, is not decided by those results. This version of the note asks that question directly.

Together with IUT IV, the step has the strength of an explicit height inequality. Every proof of such an inequality has a step of this strength, at the place where its content enters. The fact is not offered as evidence for or against the existence of the construction; it is used only as a criterion by which a supplier, once found, can be recognized. The question asks where, not how, and it can be answered by pointing to a statement or to a passage of a proof.

The step is the value-group counterpart of a compatibility that is proved on the unit side, where cyclotomic rigidity gives a unique isomorphism. On the value-group side the two profiles differ by the factors j^2 between the labels, and the counterpart cannot be an isomorphism: in that form it would force the degree scale of the Θ-link to be 1. It has to be an inclusion in the holomorphic hull, and that inclusion is the inequality. Section 1 locates where the source divides the two sides. Step (i) of the proof splits the Θ-link into a unit part and a value part, and the compatibilities of Theorem 3.11(iii) concern the unit part; we found no statement that carries the value part to regions in the common container. Since the proof of Theorem 3.11 states that its assertions follow immediately from the definitions and the cited references, the question becomes which definition or cited construction supplies that one edge. We read five places that could supply it, the family of possible images, the holomorphic hull, the identification of theta and Gaussian monoids, the compatibility statement of Theorem 3.11(iii)(c) and the global realified Frobenioids of Step (xii), as holding in the author's models of the cited statements, built from actual local objects, in which the step fails. This is a reading, not a claim that the construction does not supply the step. The question is which property of the actual construction excludes those models, that is, where the models fail to represent it, or which of the readings is wrong. The questions are not addressed to any particular person or project.

The small questions Q0–Q7 of earlier versions, on local shears, their globalization and object membership, are kept in appendices as component questions about one route, through the difference of the theta and algebraic sections of [EtTh]. This version answers them provisionally: the difference of sections does not move degree readouts, a degree-moving freedom is one positive scalar common to all places, and the route does not carry the profile switch. The abstract implication cores used by the component questions are checked in Lean, as before, and a standalone Lean file checks the elementary forms that the readings use; the readings themselves are not formalized.

No claim is made that Corollary 3.12, the ABC conjecture or the profile switch is false, or that the construction does not supply the step.

Keywords: inter-universal Teichmüller theory, Corollary 3.12, Step (xi), simultaneous holomorphic expressibility, value-group profile, local shear, mono-theta environment, anabelian geometry, Lean.

Note added in version 2.0.0. This version adds Section 1, the central question, and makes it the main question of the note; Section 1.5 locates the one edge of the comparison for which no statement was found, Section 1.6 describes the author's models, reads five places that could supply the edge as holding in them, and asks which property of the actual construction excludes them or which reading is wrong, and Appendix C gives provisional answers to Q0–Q7; a standalone Lean file checks the elementary forms that the readings use (Section 2.5). The abstract, the position statement and the closing section are rewritten accordingly; each component question receives a line on its bearing on the central question; the sentence of earlier versions saying that none of the questions is Corollary 3.12 or ABC is replaced by Remark 1; earlier works on the same step are cited ([Higuchi], [Pajo], [Takkun], [Saito], [sfingali], [Geiger], [Form], [Ri]); and the references to the author's notes are updated. The component questions Q0–Q7, Statement P′ and the related tables are moved to Appendices A–F; apart from the lines on their bearing, a paragraph on their scope and the provisional answers, their content is unchanged. Version 1.0.2 rewrote the comments of the Lean files in English and changed no question.

Position of this note

This note is a problem statement for joint examination, not a completed proof. Its questions are not addressed to any particular individual or project; they are addressed to anyone with relevant expertise who can supply a reference, a correction, a counterexample or a better formulation. Answers can be sent to the e-mail address on the title page or posted as an issue in the public repository that holds this note, `https://github.com/turahigashi/iut-explicit-initial-theta`; a one-line answer in one of the forms of Section 1.7 is enough. The note does not claim a proof or disproof of Corollary 3.12 or the ABC conjecture, a determination of H*val, or an equivalence between Statement P′ below and the residual. It is not an alternative to, or a criticism of, Project LANA's (9-1): (9-1) is one of the reference forms of the step asked about in Section 1, and the component questions separate definitions, local mechanisms and globalization steps that precede or surround it. Even if every component question receives a positive answer, the two return bridges and the numbered IUT inputs remain separate. Conversely, a concrete countermodel to one component question would locate a place where this route closes.

1 The central question

1.1 The step

Corollary 3.12 of [IUTchIII] is proved in twelve steps, and the inequality is drawn in Step (xi) [IUTchIII, pp. 181–185]. We summarize the substeps that matter here. The summary is our reading and is meant to be checked.

(xi-a) The $\Theta^{\times\boldsymbol{\mu}}_{\mathrm{LGP}}$-link from $(0,0)$ to $(1,0)$ is read as a gluing isomorphism under which the Θ-pilot object at $(0,0)$, regarded as an object of the relevant global realified Frobenioid, corresponds to the q-pilot object at $(1,0)$.

(xi-b) The multiradial algorithm of [IUTchIII, Thm. 3.11] constructs a collection of possibilities for output data in a container common to the two columns, with the properties (IPL) and (SHE) of [IUTchIII, Rem. 3.11.1(iii)]. The author states that at this step only the qualitative properties (IPL), (SHE) and (APT) of the algorithm will be used, "temporarily forgetting" that it involves the Θ-pilot, theta functions and mono-theta environments, which form its hidden internal structure in the sense of [IUTchIII, Rem. 3.11.1(iv)].

(xi-c) The possibilities for output data are related, through isomorphisms of $\mathcal{F}^{\Vdash\blacktriangleright}$-prime-strips, to the representation of the q-pilot obtained by the column-one log-Kummer correspondence (IPL), and their construction is expressible in terms of the holomorphic structure of column one alone (SHE). After passing to the holomorphic hull they are represented by localizations of arithmetic vector bundles.

(xi-d) Taking determinants of suitable positive tensor powers and procession-normalized log-volumes yields the half-line $\mathbb{R}_{\leq -|\log(\underline{\underline{\Theta}})|}$ of possible values and the number $-|\log(\underline{\underline{q}})|$ attached to the q-pilot. The column-one log-Kummer correspondence is used here to correct the vertical misalignment of the unit parts.

(xi-e) (SHE), that is, expressibility in terms of the arithmetic holomorphic structure of column one, is read as expressibility by operations that remain valid when the input prime-strip is subject to the condition that its pilot log-volume equals the fixed value $-|\log(\underline{\underline{q}})|$.

(xi-f) Hence the construction of the possible values is also, perhaps only up to a kind of approximation caused by the indeterminacies, a construction of $-|\log(\underline{\underline{q}})|$, so that $-|\log(\underline{\underline{q}})| \in \mathbb{R}_{\leq -|\log(\underline{\underline{\Theta}})|}$. This is the inequality.

(xi-g) The argument is summarized as two tautologically equivalent ways of computing the log-volume of the q-pilot [IUTchIII, Fig. 3.8].

The question of this note concerns the passage from (xi-e) to (xi-f): what supplies (SHE) in the form used in (xi-e).

1.2 What is already known

(K1) *The step against the cited statements.* At the level of log-volumes, the form of (SHE) in (xi-e) is the conclusion rewritten; it is equivalent to the inequality [U26, Prop. 6]. Its weak reading, that the possible images are computed from the common container, follows from the statements that the proof cites, taken at the strength of the source [U26, Prop. 5]. The strong readings, and the equality (9-1) that the interim report of Project LANA identifies

as the missing step [LANA, §9.2], do not [U26, Thm. 10, Cor. 11]. One of the models that show this is built from actual local objects at a finite extension of $\mathbb{Q}_p$ carrying the local data of a theta function: tensor packets of the field with the mono-analytic integral structure, the indeterminacies (Ind1) and (Ind2) acting through all permutations of tensor factors and all automorphisms of the log-shells, values of the theta series, a container containing all horizontal and vertical images, and, as hull, the smallest multiple of the ring of integers of the packet that contains the container [U26, Thm. 22]. The models survive the addition of any finite set of further places [U26, Thm. 14], and the closed loop of [IUTchIII, Rem. 3.9.5(viii)(ix)], to which [IUTchIII, Rem. 3.11.1(iii)] relates (SHE), holds in them.

(K2) *The step in a conditional assembly.* In [U26c] the chart calibration, the normalized-generator scale and the unit side are supplied at the level of stated inputs, and the remaining input is the value-group profile switch H*val: from the $\{j^2u\}_j$-type profile of the Θ-pilot, an actual q-pilot representation compatible with the holomorphic hull is obtained [U26c, §8]. The model of [U26d], with the source profile at $\ell = 7$, satisfies every other input of the assembly and falsifies the conclusion. In the language of [U26b], the step is a failure of completeness at an interface, and the question becomes whether the additional principle needed is supplied by the actual construction.

(K3) *Earlier work on the same passage.* Higuchi [Higuchi] proves non-derivability for a qualitative package extracted from (xi-e) and (xi-f), and Pajo [Pajo] a no-go theorem for a scale-free monoid, hull and log-volume abstraction. In Lean, Takkun-kohinata [Takkun] proves for its transcription of the passage that the volume bound of the multiradial representation together with the containment is equivalent to the height inequality, that strict identification of the copies, with volume-preserving indeterminacies and a hull no larger than the Θ-pilot, is contradictory, and that the slack between the hull and the Θ-pilot cannot come from the orbits of (Ind1) and (Ind2), only from the hull. Saito [Saito] audits the passage from (xi-a)–(xi-e) to the membership of (xi-f) against a registered inventory of the source; sfingali [sfingali] formalizes in Lean the core of the objection of [SS]; and Geiger [Geiger] records that the decisive membership of (xi-f) is stated to follow formally from a pair of declared properties rather than derived. The public Lean repository of Project LANA [lana] proves that Step (xi) cannot be derived from a package collecting what the repository proves at the level of Theorem 3.11, shows that Step (xi) adds to the package exactly the inequality $(s-1)a \le V$ in its notation (`xi_iff`), and separates two readings of (SHE), of which the one asserting that some output possibility has the volume class of the q-pilot region is proved equivalent to the wall inequality of [LANA, §0.2] (`sheq_iff_wall`). The interim report of the project states that the project does not at present have a proof of (9-1) [LANA, §10.5].

(K4) *The author of IUT on the same step.* In his report on formalization [Form, §4], Mochizuki reorganizes the implication from Theorem 3.11 to Corollary 3.12 into four commutative triangles and isolates an intermediate statement "3.11.5" (Theorem 3.11 together with Remark 3.9.5), which, as the report notes, does not appear in [IUTchI]–[IUTchIV]. The skeletal Lean code described there for "3.11.5 $\Rightarrow$ 3.12" takes as an input an arrow (SHE), which takes the input data of the algorithm to be the data of the q-pilot constructed from the étale Hodge theatre. The report also states that the implication from Theorem 3.11 to Corollary 3.12, which has attracted the most attention, is in fact not very difficult, and that, correctly understood, it shows that the real object of interest is the proof of Theorem 3.11 [Form, §2]. At the level of log-volumes the arrow (SHE) is the carrier reading of [U26, Def. 4] [U26, §10]. Richards' Lean port [Ri] states "3.11.5" as a proposition and derives it, in a separate file, from a structure

with a field identifying the region of the Θ-pilot with that of the q-pilot.

1.3 What the step must deliver

We state it in terms of local fields, log-shells and Haar measure. Fix a number field K, an elliptic curve E/K, a prime $\ell \geq 5$ and the remaining data of an initial Θ-datum [IUTchI, Def. 3.1]; further notation is that of Appendix A. At a bad place v put $u_v = |\log(\underline{\underline{q}}_v)| = (2\ell)^{-1}\log(\mathfrak{q}_v)$ [IUTchIV, Thm. 1.10]. The labels are $j = 1, \ldots, \ell^* = (\ell-1)/2$. At v the Θ-pilot has the labelled log-volume profile $\{-j^2 u_v\}_j$ and the q-pilot the profile $\{-u_v\}_j$ [IUTchII, Cor. 4.5(iv)(v)], [IUTchIII, Def. 3.8(i)]. The procession normalization gives every capsule the same weight [IUTchIII, Prop. 3.9(i), Rem. 3.9.3], and the aggregates of the two profiles are $-\frac{\ell(\ell+1)}{12}u_v$ and $-u_v$ [U26d, §3].

The possible images of the Θ-pilot are its images under the indeterminacies of [IUTchIII, Thm. 3.11]: (Ind1), coming from automorphisms of the étale-like data (copies of the local absolute Galois groups), which act on log-shells through mono-anabelian reconstruction; (Ind2), coming from automorphisms of the unit parts; and (Ind3), the upper semi-compatibility of the log-Kummer correspondence [IUTchIII, Thm. 3.11(ii)]. The holomorphic hull of their union is formed in tensor packets of log-shells, and volumes are procession-normalized log-volumes with respect to Haar measure [IUTchIII, Rem. 3.9.5(vii)]. Explicit formulas for (Ind1)–(Ind3) in terms of local fields are given in [DHPS].

The step must deliver the following.

(S) The region of the q-pilot, represented in the common container through the column-one log-Kummer correspondence, lies in the holomorphic hull of the possible images of the Θ-pilot (*carrier form*); or at least its log-volume does not exceed that of the hull (*value form*):

$$-|\log(\underline{\underline{q}})| \;\leq\; -|\log(\underline{\underline{\Theta}})|.$$

The value form is the inequality of Corollary 3.12. In the reading of [U26, Def. 4], the carrier form implies (9-1), and (9-1) implies the value form [U26, Prop. 6]. Together with the upper bound for $-|\log(\underline{\underline{\Theta}})|$ in [IUTchIV, Thm. 1.10], (S) for all initial Θ-data gives, through [IUTchIV, §2], the diophantine inequalities of [IUTchIV, Cor. 2.3].

1.4 What is known not to suffice

(N1) *The log-volume content of the cited statements.* The 22 statements that the proof of Corollary 3.12 uses as definitions or premises, among 63 cited targets [U26, §3, App. A], include what [IUTchIII, Thm. 3.11] states about the output of the algorithm, the hull of [IUTchIII, Rem. 3.9.5], (IPL) and the closed loop. Their content at the level of log-volumes holds in the models of (K1), described in Section 1.6, in which (S) fails. So does the weak reading of (SHE).

(N2) *Scalar calibrations.* Agreement of the measuring charts of the two routes [U26c, Lemma 8.1], agreement at the normalized generator $j = 1$, and compatibility on the unit side [U26c, Lemma 8.2] hold in the model of [U26d], in which (S) fails. A separate assumption that the aggregate ratio of the two pilots equals 1 contradicts the source readout $\ell(\ell+1)/12 \neq 1$ [U26d, §7.5].

(N3) *Identification of the two pilots.* A statement identifying the image of the Θ-pilot with the region of the q-pilot implies (S), but it forces the degree scale of the Θ-link to be 1 [U26,

Prop. 26(b)], contrary to the source readout; with the source data it cannot hold. Compare [SS], [sfingali] and `strict_collapse` in [Takkun].

In the language of [SL], a statement of type (N3) asks for a strict 2-cell between the two paths, which exists only if the degree scale is 1 [SL, Thm. 5.2]; (S) is a lax 2-cell [SL, Thm. 5.3]; and the properties (IPL), (SHE) and (APT) by themselves are compatible with any size of the blurring, so that neither (S) nor its negation follows from them [SL, Thm. 5.7].

1.5 The value-group counterpart, and the edge without a statement

The step is the value-group counterpart of a compatibility that is proved on the unit side. On the unit side the indeterminacies (Ind1) and (Ind2) preserve log-volumes [IUTchIII, Cor. 3.12, proof, Step (x)], and what is needed is an isomorphism, made unique by cyclotomic rigidity [EtTh, Cor. 2.19(i)]. In the conditional assembly this side is supplied by such uniqueness statements [U26c, Lemma 8.2], and [EtTh, Cor. 2.19(i)] is also the solved form from which Statement P′ departs (Appendix D). On the value-group side the two profiles differ by the factors j^2 between the labels. A counterpart of the same type, an isomorphism compatible with log-volumes, would identify the two pilots and force the degree scale of the Θ-link to be 1, as in (N3). What is needed is an inclusion in the hull, (S): a lax rather than a strict 2-cell [SL, Thms. 5.2, 5.3]. This is why the step does not follow by analogy from the unit side, and why it carries the strength of the conclusion.

In the column-one container the proof compares what is carried along two routes.

Route 1 The Θ-pilot at $(0,0)$, the value part of the $\Theta^{\times\boldsymbol{\mu}}_{\mathrm{LGP}}$-link, the q-pilot at $(1,0)$, and the Kummer isomorphisms of column one [IUTchIII, Thm. 3.11(ii)]. The statement of Corollary 3.12 defines $-|\log(\underline{\underline{q}})|$ as the log-volume of this image, computed in the same multiradial representation and not subject to (Ind1)–(Ind3) [IUTchIII, p. 174].

Route 2 The Θ-pilot at $(0,0)$, the Kummer isomorphisms of column zero, and the permutation symmetry at the end of [IUTchIII, Thm. 3.11(i)], which carries the possible images into the column-one container (Step (xi-b)).

Step (i) of the proof splits the Θ-link into a unit part, a full poly-isomorphism of $\mathcal{F}^{\vdash\times\boldsymbol{\mu}}$-prime-strips, and a value part, a full poly-isomorphism of $\mathcal{F}^{\vdash\blacktriangleright}$-prime-strips that sends the Θ-pilot to the q-pilot [IUTchIII, p. 175]. The edges of the comparison and the statements that supply them are as follows.

Edge	Statement
Kummer isomorphisms of both columns	[IUTchIII, Thm. 3.11(ii)]
Permutation symmetry between the columns	[IUTchIII, Thm. 3.11(i)], end
Compatibility with the unit part of the Θ-link	[IUTchIII, Thm. 3.11(iii)(a)(b)]
Shielding of the cyclotomes that give the value part	[IUTchIII, Thm. 3.11(iii)(c)(d)]; Step (ii) of the proof
Log-volumes, their compatibility with log-Kummer, the hull	[IUTchIII, Thm. 3.11(ii), end; Prop. 3.9(iv); Rem. 3.9.5]
Value part of the Θ-link, carried to regions in the container	none located

The compatibilities of [IUTchIII, Thm. 3.11(iii)] are stated with the full poly-isomorphisms of $\mathcal{F}^{\vdash\times\boldsymbol{\mu}}$-prime-strips as the example, in the sense that the constructions are stable, equivariant and functorial with respect to arbitrary automorphisms of the domain and the codomain of the Θ-links

[IUTchIII, pp. 157–158]. Step (xi-b) describes the relation between the value parts as consisting precisely of full poly-isomorphisms of $\mathcal{F}^{\vdash\blacktriangleright}$-prime-strips, and Step (xi-e) crosses the remaining edge by its reading of (SHE). The proof of [IUTchIII, Thm. 3.11] states that each assertion follows immediately from the definitions and the references cited in it [IUTchIII, p. 159]. So if something supplies the edge, it lies in those definitions and cited constructions, or in the hidden internal structure that Step (xi-b) sets aside. This is our reading of the statements, and the question below asks for it to be confirmed or corrected.

1.6 The models, the test, and the question

The models. The models used in this section are the author's own, from [U26] and [U26d]. They are countermodels to a derivation: the statements in question hold in them, and (S) fails. They are not counterexamples to Corollary 3.12, and they are not actual global initial Θ-data.

(M1) *The local model of (K1)* [U26, Def. 20, Thm. 22]. Let L be a finite extension of $\mathbb{Q}_p$ that carries a theta datum: $\tau, i \in L^\times$ with $i^2 = -1$ and $|\tau| < 1$, elements ξ_1, ξ_2, ξ_3 with $\xi_j^7 = \vartheta(i)/\vartheta(\pm\tau^j i)$ for the theta series ϑ of [EtTh, Prop. 1.4], and r with $r^{14} = \tau^2$. The Tate parameter is $q = \tau^2$, so that $\xi_j^{14} = q^{j^2}$ ($\ell = 7$, labels $j = 1, 2, 3$); for every $N \geq 1$ the N-th powers again form a theta datum, with Tate parameter q^N. At the stage j the packet is the tensor product P_j of $j+1$ copies of L over $\mathbb{Q}_p$, with ring of integers $\widetilde{\mathcal{O}}$ and log-volumes normalized as in [IUTchIV, Prop. 1.4(i)]. The Θ-pilot is $\xi_j^N \cdot \widetilde{\mathcal{O}}$ and the q-pilot is $r^N \cdot \widetilde{\mathcal{O}}$. The possible images are all regions between a horizontal image and the container. The horizontal images are the images of the Θ-pilot under all $\mathbb{Q}_p$-linear automorphisms of P_j that preserve the log-shell, all unit multiples and all theta data with the same τ, i; the container is the p-power multiple of the log-shell packet that contains all horizontal and vertical images; and the hull is the smallest multiple of $\widetilde{\mathcal{O}}$ that contains the container. The excess of the hull over the Θ-pilot is at most a constant E_L that depends only on L, whereas (S) requires $\frac{11}{42} N \,|\log|q|| \leq E_L$; so (S) fails for large N. The model satisfies the 22 cited statements that carry log-volume content, and survives the addition of any finite set of further places [U26, Thms. 10, 14, 22]. Every choice in it that the source does not dictate makes the hull larger, and so works against the failure of (S) [U26, Rem. 23].

(M2) *The profile countermodel* [U26d]. At $\ell = 7$, with labels $j = 1, 2, 3$ at one bad place and equal weights $1/3$, the Θ-side readouts are $-j^2$ and the q-side readouts are -1, as in the source. The normalized generator, the unit side, membership, monotonicity, the hull domain and both volume decompositions hold, while the aggregates are $-14/3$ and -1, and the conclusion of the shape of Corollary 3.12 fails.

Both are given in finite, machine-checked form [U26, §11], [U26d, §6]. The question of this section is, in other words, where these models fail to represent the actual construction.

The test. Call a statement P, that is, a numbered statement or a passage of a proof, in [IUTchI]–[IUTchIV] or in the papers they cite, a *candidate supplier* if it passes the following tests.

(T1) *It does work.* Some consequence of P for regions, log-volumes and labels fails in (M1) or in (M2). Otherwise P falls under (N1) or (N2).

(T2) *It keeps the profile.* P is consistent with the source readout: it does not force the aggregate ratio $\ell(\ell+1)/12$ of the two pilots to equal 1. Otherwise P falls under (N3).

(T3) *It is not the conclusion.* P is established without using Step (xi-f) or Corollary 3.12.

Because (M1) is built from actual local objects and survives the addition of places, a statement passing (T1) must use something that the cited statements do not state at the level of log-volumes: for instance a property of the hull or of the family of possible images beyond those stated [U26d, §7.6], global information about the initial Θ-datum, or the structure that Step (xi-b) sets aside. A candidate that holds in (M1) and (M2) is not a supplier, whatever else it does. The tests are necessary conditions, not sufficient ones.

Central question. Five places in the source could supply the edge of Section 1.5. We read each of them as holding in (M1) and (M2), in which (S) fails. This is our reading of the statements, not a claim that the construction does not supply the step.

(a) *The family of possible images* [IUTchIII, Thm. 3.11(i); Props. 3.1–3.10]. (M1) reads it maximally, as all regions between a horizontal image and the container [U26, Def. 20]; the actual family lies in it.

(b) *The holomorphic hull* [IUTchIII, Rem. 3.9.5(vii)–(x)]. The hull of (M1) is the hull of the whole container; the actual hull, the hull of the possible images [IUTchIII, Rem. 3.9.5(i)], lies in it. Even so, its excess over the Θ-pilot is at most E_L, while the excess that (S) requires grows linearly with N [U26, Thm. 22(a)].

(c) *The identification of theta and Gaussian monoids* [IUTchII, Cors. 4.5(iv)(v), 4.6(iv)(v)]. Its evaluation isomorphisms, and the ratio vector $(\dots, j^2, \dots)$, are constructed inside one Hodge theatre on the Θ-side, and the q-pilot does not appear in them; the Θ-pilot of (M1) is already built from the Gaussian values.

(d) *The compatibility statement* [IUTchIII, Thm. 3.11(iii)(c)]. Its data are those of [IUTchIII, Cor. 2.3]: a Hodge theatre, the $\mathcal{F}^{\vdash}$-prime-strip of the environment, the data at the bad places, the unit part $\mathcal{F}_{\triangle}^{\vdash\times\boldsymbol{\mu}}$ and a full poly-isomorphism of unit parts. The multiradiality of [IUTchIII, Cor. 2.3(i)] is that the functor from these data to the coric data $(\mathcal{D}^{\vdash}, \mathcal{F}^{\vdash\times\boldsymbol{\mu}}(\mathcal{D}^{\vdash}))$ is full and essentially surjective [IUTchIII, pp. 72–75]. The value part $\mathcal{F}_{\triangle}^{\vdash\blacktriangleright}$, which carries the q-pilot, is in neither. At the level of regions its content is the closure under (Ind1) and (Ind2) and the bound (Ind3) [U26, §9], which hold in (M1); and it does not involve the value of the q-pilot.

(e) *The global realified Frobenioids* [IUTchIII, Cor. 3.12, proof, Step (xii), p. 185]. Their role, as stated there, is to fix whether a local object corresponds to a given open submodule of the log-shell or to a p^N-multiple of it, since the isomorphisms of Frobenioids that arise from the Θ-link preserve only isomorphism classes of objects [IUTchIII, Rem. 3.6.2(i)]. In (M1) the regions and the container are given subsets of the packet, at a fixed scale, so this ambiguity does not arise. The freedom that remains in the degree is one positive scalar common to all places (Q0, Q1), which acts on all labels alike and so cannot carry one profile to the other.

These readings depend on the container being a p-power multiple of the log-shell packet [U26, §6]. If they are right, (M1) and (M2) satisfy (a)–(e). Which property of the actual construction excludes them, and where is it stated or constructed? In other words, where do these models fail to represent the actual construction? Or which of the readings (a)–(e) is wrong?

Our readings are least certain at the following places, and a pointer on any one of them would be enough.

(i) What is the internal structure of the value parts that Step (xi-b) sets aside, and does it constrain regions?

(ii) Is the container a p-power multiple of the log-shell packet, or is it larger, for instance growing with N?

(iii) Does the family of possible images constructed in [IUTchIII, Props. 3.1–3.10] lie within the family of (M1), that is, between a horizontal image and the container?

(iv) Do the product formula and the global realified Frobenioids do more than leave one positive scalar common to all places?

(v) Does a categorical statement, for instance [IUTchIII, Cor. 2.3(i)] or [IUTchIII, Thm. 3.11(iii)], have a consequence for regions or log-volumes that (M1) violates?

The provisional answers in Appendix C also remove the difference of the theta and algebraic sections of [EtTh] as a candidate (Q2, Q6).

In the decomposition of [Form, §4], this asks where the arrow (SHE), which the skeletal code takes as an input, is supplied. In the conditional assembly of [U26c], the carrier form of (S) is the content of H*val; in the reading of [U26, Def. 4] it implies the equality (9-1) of [LANA, §9.2], which in turn implies the value form [U26, Prop. 6].

Among the statements that the proof of Corollary 3.12 cites, none passes (T1) in the reading of [U26]: the 22 that carry log-volume content hold in its models [U26, Thms. 10, 22], and the others are discussed in [U26, §9]. This concerns the statements as they are cited; what their proofs construct, in particular the proof of Theorem 3.11, is what the central question asks about. Other places in the source are listed in Appendix G.

1.7 Forms of answer

Any of the following is useful, and one line is enough.

(A) *Misread.* A reading above is wrong: one of (a)–(e) does not hold in the models, because the actual construction is larger there (where); or (M1) or (M2) violates a cited statement (which one); or Step (xi-e) does not use (SHE) in the form described in Section 1.1 (in which form, then).

(B) *Located.* A property of the actual construction, not among (a)–(e), that excludes (M1) or (M2), with a page, together with one consequence for regions, log-volumes or labels that fails in that model.

(C) *Not needed.* The argument reaches (S) by a route other than (xi-e)–(xi-f); please name the route.

(D) *Not yet written.* The supplier is contained in the construction but not yet stated in a citable form; please indicate the part of the construction, for instance a part of the proof of Theorem 3.11 or one of the triangles of [Form, §4].

1.8 Strength is not evidence

Remark 1. A candidate supplier has, together with [IUTchIV, Thm. 1.10], the strength of an explicit height inequality, because (S) has. This is a property of every proof of such an inequality: at the step where its content enters, a proof carries the strength of its conclusion. The fact is therefore expected whether or not the construction supplies the step. It is not offered as evidence either way, and it does not make the question hard to answer, since the question asks where and

not how. It is used above only as a criterion: a supplier, once found, can be recognized by what it implies. The component questions of Appendix C are weaker than the central question, and each informs a part of it.

2 Fixed background

2.1 The residual in two-path form

The residual in the conditional assembly [U26c] compares two routes at the level of objects. Route 1 starts from the Θ-pilot, passes through the value-group part of one Θ_{LGP}-link to the q-pilot, and then uses the column-one log-Kummer correspondence. Route 2 starts from the same Θ-pilot, uses the column-zero log-Kummer correspondence and permutation symmetry to obtain a possible image. The object-level assertion (α) says that the Route 1 representation is, up to the permitted indeterminacies, one of the Route 2 possible images. The numerical assertion (β) follows from (α) together with the decomposition identity expressing the q-side readout as the volume of that object and monotonicity of the hull volume. We use (α) as the standard form of the residual.

2.2 Parts treated as supplied

The questions below fix the following layers as supplied at assembly-statement level. This is a focusing device, not a claim that they have been proved unconditionally in the actual theory.

Layer	Content	Status here
Chart	Calibration, generator scale and transport distortion of the measuring chart	Treated as supplied
Unit	Theorem 3.11(iii)(a)(b), Kummer classes of actual functions and the restriction bridge	Treated as supplied
Semantics	Reading prime-strip morphisms as isomorphisms of collections of data	Fixed as a reading; actual inhabitation remains separate
Profile	Hull compatibility of the switch $\{j^2u\}_j$ to $\{u\}_j$	**Open; subject of this note**

2.3 What the profile countermodel fixed

The countermodel of [U26d] uses concrete data at $\ell = 7$ in which chart calibration, the normalized generator, the unit side, membership and both volume decompositions hold, while only the labelwise profile differs and the Corollary-3.12-shaped conclusion fails. The aggregate scale is in general $12/(\ell(\ell+1))$. Hence an input solving the residual cannot be merely the numerical assumption that an aggregate scale equals 1; it must connect profiles at object level or at the level of a sufficiently rich readout.

2.4 Nearby public Lean work

The public Lean development of Project LANA [lana] already contains formalizations close in theme, including abstract forms of [EtTh] Proposition 2.14, shift automorphisms, discrete and cyclotomic rigidity, and, in Wall/LogLink models, scaling and unit-induced properties, a profile coefficient `sFactor`, and hexagon noncommutativity. This note does not restate those developments. Its questions concern how such model-level classifications connect with the categories of the source texts and with globalization of section differences.

2.5 Formal cores checked in Lean, and what was not checked

Before circulation, the parts of the implication map that can be stated without the [EtTh] or IUT constructions were isolated in the Lean module:

`M3.relative.ActualValuePartQuestionImplications`.

A selected build over three modules passed, and `#print axioms` for the main theorems used only the standard three axioms. The checked abstract implications are:

- two splittings of one group homomorphism agree on the quotient, their pointwise difference lies in the kernel and satisfies the usual cocycle identity; hence any readout factoring through that quotient agrees for the two splittings;
- object membership, the readout decomposition identity, the hull-domain condition and monotonicity imply the numerical volume bound, abstractly $(\alpha) \Rightarrow (\beta)$;
- equality of an otherwise unconstrained transported number with the q-side readout does not imply that bound, and equal totals do not determine a labelled profile;
- a normalized linear readout invariant under all label permutations is an average, the mean-square formula holds, and a common ratio $r \neq 1$ at every place does not cancel in a nonzero total.

For this version, the elementary forms that the readings (a)–(e) of Section 1.6 use were checked in a standalone file that depends only on Mathlib,

`CentralQuestionReadings.lean`,

compiled with Lean 4.30.0; for each of its 19 theorems, `#print axioms` shows only the standard three axioms. It checks the following.

- Readings (a) and (b): an upper bound for the hull of the maximal family is an upper bound for every smaller family; the hull of a family that contains the container and lies in it is the hull of the container; and an excess bounded independently of N fails, from some N on, a requirement that grows linearly in N.
- Readings (c) and (d): a statement about data that do not contain the value of the q-pilot also holds in the configuration that changes only that value, where (S) at the level of values fails; so such a statement does not imply (S).
- Reading (e), with Q0, Q1, Q2 and Q6: a scaling of the local degrees that keeps the families of total degree zero is one scalar common to all places; one scalar acting on all labels alike does not carry the profile $\{a\}_j$ to $\{j^2 a\}_j$, nor back, once two labels are present; and neither does a transport that keeps every degree readout.

Examples in the same file show that the positive slope, the condition that the members lie in the container, the two labels and the second place are needed, and removing any one of six hypotheses makes the corresponding proof fail. These are the forms that the readings use. That the constructions of [IUTchIII] and the models of [U26] have these forms is the content of the readings, and it is not formalized.

The Lean files do not formalize the readings of Section 1.6 themselves, mono-theta environments, the actual difference δ, its globalization, Statement P$'$, the two return bridges, or dictionary-dependent relations among candidate forms. They should not be read as verifying the questions themselves.

2.6 Relation to prior public claims, in the audited range

The central identification difficulty itself is not new: it has been a public issue since [SS], and Project LANA records a corresponding membership problem as (9-1) [LANA, §9.2]. Earlier work on the same passage is listed in Section 1.2. Joshi has published, inside his own arithmetic Teichmüller theory, a claimed local prototype of Corollary 3.12 and valuation-scaling results [JosII]. Because the definitions and framework differ, this note does not treat those claims, whose evaluation is not attempted here, as directly closing Questions Q1 or Q2 as phrased in the [EtTh] categories. For local outer automorphisms arising from the Jannsen–Wingberg presentation, see [NSW], [HN] and [Kon]; in the statements audited here, the emphasis is on unit/log-side actions, not a category-by-category classification of actions on a normalized degree quotient or a one-place global realization. A public document [Zen26] claims an additional hull-inflation obstruction even under (9-1); that claim is not verified or used here. Accordingly, this note does not claim that Q0–Q7 are globally unprecedented or open in every formulation. In the range audited, no numbered statement was located that closes them in the exact categories and with the exact readouts used below.

3 Closing: what guidance would be decisive?

The provisional answers in Appendix C settle the component questions as far as the cited statements allow. Along the route through the difference of sections, global synchronization leaves only one positive scalar in the degree (Q0, Q1), the difference of sections does not move degree readouts (Q2, Q5), and the two-path difference is not that difference (Q6).

Any of the following would be decisive for this route:

- for the central question, any answer of the forms (A)–(D) of Section 1.7, in particular a pointer on one of the places (i)–(v) of Section 1.6;
- for the component questions, a correction of the provisional answers in Appendix C, in particular those to Q0(4), Q1, Q2 and Q6, or a reference that settles them;
- a prior public formulation of the residual in [EtTh] or anabelian reconstruction vocabulary. This note does not make its value depend on a novelty claim.

The central question concerns a step that carries the strength of the conclusion. As Remark 1 explains, this is expected of the step at which the content of any proof of the inequality enters, and it does not make the question hard to answer. The component questions are weaker, and each informs a part of the central one. The earlier works of Section 1.2, the conditional assembly and the profile countermodel have localized the remaining difficulty to that step, and Section 1.5 locates it at one edge of the comparison: the value part of the Θ-link, carried to regions in the common container. The purpose of this note is to make the step searchable, and to put its parts in a form that specialists can judge in everyday vocabulary.

Residual	Content	Treatment here
R1	Numbered statements of IUT I–IV as black boxes	Not verified
R2V	Value-group profile switch H*val	Located at one edge (§1.5) and asked in §1.6; component questions answered provisionally (App. C)
R2	Inhabitation / frame fidelity to the actual construction	Marked at Q6 and Form B seams
R3	Source checking and independent reverification	Author checking and outside rereading remain

A Setup and terminology

Appendices A–F keep the component questions of earlier versions together with the setup that they use, with provisional answers. They are not needed to state or to answer the central question of Section 1. Appendix G lists further places in the source.

A.1 Stage

Let K be a number field, E/K an elliptic curve, and S a finite set of places at which E has split multiplicative reduction. For $v \in S$, let $q_v \in K_v^\times$ be the Tate parameter, with $\mathrm{ord}_v(q_v) > 0$ for the normalized valuation. Fix a prime $\ell \geq 5$, a bound inherited as a hypothesis from the assembly-side data conditions. For each v, consider the Tate curve, its ℓ-torsion points, the relevant hyperbolic fundamental groups, and decomposition groups at cusps and torsion points. Labels are $j \in \{1, \ldots, \ell^*\}$, $\ell^* = (\ell - 1)/2$. Write $\eta_{v,j}$ for the Kummer class of the theta value. Its valuation satisfies

$$\mathrm{ord}_v(\theta_{v,j}) = \frac{j^2}{2\ell}\,\mathrm{ord}_v(q_v).$$

With $u_v := |\log(\underline{\underline{q}}_v)| = (2\ell)^{-1}\log(\mathfrak{q}_v)$, the signed readouts are $-j^2 u_v$ on the theta side and $-u_v$ on the q side.

A.2 Two sections and their difference

A mono-theta environment in [EtTh] contains tempered fundamental group data and a theta section. The cyclotomic envelope also has a tautological algebraic section. In a relevant subquotient the two appear as splittings of one surjection, with the theta section obtained from the algebraic section by subtracting an η-cocycle. We write δ generically for such a difference or for the change of section under a change of model, always naming the category in question. This avoids confusing the standard η-cocycle of [EtTh] with indeterminacy under conjugation or replacement of models.

A mono-theta environment contains the theta section as data; the algebraic section is detected, in subquotients, through commutators. A bi-theta environment contains both sections but loses discrete rigidity. This tension is Question Q3.

A.3 What "shear" means here

Because the word is used with different scopes, we use a working definition. For a replacement α of splittings, models or identifications in a specified category, fixing specified group-theoretic and cyclotomic data, consider the induced action on degree readouts. Conceptually there are two types:

Type	Form of action	Effect on degree
U (splitting shear)	Splitting changes by a unit-valued cocycle	Normally invisible in the valuation quotient
D (degree shear)	Normalized degree readout itself changes non-trivially	If linear, $D \mapsto \lambda D$, $\lambda \neq 1$, is the issue

A warning is essential. Local automorphisms arising from the Jannsen–Wingberg presentation and often called shears mix generators and induce unipotent actions on the unit/log side [NSW, Thm. 7.5.14], [HN], [Kon, Thm. 1.1]. Under suitable hypotheses such an action need not be a scalar composed with a field automorphism, and it may move sizes inside a log shell. This should not be identified with a D-type action on the normalized valuation quotient or degree readout itself. Q0 therefore does not presuppose the existence of D-type shears; it asks for their classification.

When an action on a generator readout is multiplicative, write its ratio as λ_v. Otherwise λ_v is only shorthand and the action itself should be recorded. For transported readouts write $\varepsilon_v := D_v^{\mathrm{tr}}(q_v) - D_v(q_v)$.

B The local skeleton already in [EtTh]

This section records a starting point, not a new result. The theta functional equation [EtTh, Prop. 1.4(ii)] and its cohomological form [EtTh, Prop. 1.5(iii)] are the source of the j^2 profile. By [EtTh, Thm. 1.6(ii)], a tempered isomorphism preserves the kernel in the valuation chain and the element $1 \in \widehat{\mathbb{Z}}$; by (iii), the $\mathcal{O}^\times \cdot \eta^\Theta$ class is preserved up to conjugation. Thus the indeterminacy of the theta class is compressed to the unit side, while its value part is pinned. Corollaries 2.8(iii) and 2.9 preserve the glued class without a constant indeterminacy and preserve cusp labels. Proposition 2.12 detects the algebraic splitting through commutators; Proposition 2.14(ii) gives a shift automorphism inducing identity on both quotient and kernel; and Corollary 2.19(i) obtains a cyclotomic isomorphism from the difference of two splittings.

Formally, a readout factoring through the common quotient takes the same value for the two splittings. This abstract part is among the Lean-checked cores. It is only readout invariance, not object identity, and it does not include identification of the actual degree readout with such a quotient readout, globalization, or transport to the procession. The questions below separate those outer steps.

C Component questions Q0–Q7

Each of the following questions is weaker than the central question of Section 1, and none is needed to state it. They concern one route, through the difference of the theta and algebraic sections of [EtTh], and they work at the level of the degree at each place. An action on the degree line of a place changes the readouts of all labels by the same factor, so it cannot by itself turn the profile $\{j^2 u\}_j$ into $\{u\}_j$. The component questions can therefore close this route, or keep it open, but they cannot by themselves supply (S); the profile enters at Q7, whose distinction between readouts and membership is now part of (S). Each question is followed by a provisional answer. An answer marked *from the cited statements* rests on the statements it cites; one marked *author's reading* also rests on our reading of how they apply, and corrections are welcome. Taken together, the answers close this route. An answer to one of them would inform a specific part of the central question, recorded under "Bearing on the central question". For each, a known reference, a countermodel, a correction of category, or a better formulation would serve the purpose of this note. Structurally,

Q0–Q5 cut out prerequisites, definitions and the local core of Statement P′ in Appendix D. Q6 and Q7 are bridges back from P′ to the assembly. Thus P′ is not an additional independent question beside Q0–Q7; it is a composite form built from Q0–Q5, with Q6–Q7 outside it.

We use the following labels for relations with (9-1): *directly connected* under a fixed dictionary; *connected under an additional identification*; *requires separate examination*; and *corresponding*, meaning essentially the same content in different vocabulary under a fixed dictionary. Here "dictionary" means a fixed correspondence between the assembly-side two paths, possible images and degree data and the membership formulation used by LANA. The dictionary itself is not proved in this note.

Q0. In which category do local shears live?

For a mono-theta environment over a local field and a degree readout D, classify the actions on D induced by automorphisms or changes of model fixing specified group-theoretic and cyclotomic data. In particular, in which of the following categories does a D-type action, moving degree itself, occur?

1. topological-group automorphisms of tempered fundamental groups, modulo inner automorphisms;
2. automorphisms of mono-theta environments;
3. automorphisms of bi-theta environments;
4. automorphisms of tempered Frobenioids and their splitting monoids;
5. automorphisms of realified Frobenioids.

Please also distinguish whether "fixing cyclotomic data" means identity action on the cyclotomic object or compatibility with its natural cyclotomic isomorphism.

Relation to (9-1): requires separate examination. (9-1) concerns membership of specific standard objects and does not classify automorphisms category by category. Even if (9-1) is true, degree-moving shears might exist in a wider category; even if they exist, the standard transport might avoid them.

Bearing on the central question. A degree-moving action acts on the degree line of a place, so it multiplies the readouts of all labels by the same factor and cannot by itself turn $\{j^2u\}_j$ into $\{u\}_j$. Its bearing is on whether the indeterminacies contain a freedom in the degree, that is, on how large the blurring is. A classification excluding such actions in a category removes that category from consideration.

Answer (from the cited statements, except item 4). An isomorphism of tempered fundamental groups induces isomorphisms of the relevant cohomology groups that are compatible with the surjections onto $\widehat{\mathbb{Z}}$ determined by the valuations, preserving both their kernels and the element $1 \in \widehat{\mathbb{Z}}$ [EtTh, Thm. 1.6(ii)]; so no D-type action occurs in category 1. An automorphism of a mono-theta or a bi-theta environment determines an automorphism of the tempered fundamental group [EtTh, Prop. 2.14(iii)], so none occurs in categories 2 and 3 either. These automorphisms act on the labels, through $(l\cdot\mathbb{Z})\rtimes\{\pm1\}$ in the mono-theta case and through a subgroup of $(N^{\dagger}l\cdot\mathbb{Z})\rtimes\{\pm1\}$ containing $(Nl\cdot\mathbb{Z})\rtimes\{\pm1\}$ in the bi-theta case. After realification the degree is determined only up to $\mathbb{R}_{>0}$ [FrdI, Thm. 6.4(ii)], so a D-type action occurs in category 5, by one positive scalar; further structure [FrdI, Thm. 6.4(iii)(iv)] reduces it to $\mathbb{Q}_{>0}$ or to 1. Category 4 is not settled by the statements read here. The two meanings of fixing cyclotomic data give the same answers, since the degree is read through the valuation surjection of [EtTh, Thm. 1.6(ii)].

Q1. Can a local shear globalize? (Main component question)

Fix the local shear objects as in Q0. What is the image, inside the product of local shears over places, of differences arising from a global change of model? When the degree action can be written by ratios $(\lambda_v)_{v\in S}$, can the family

$$\lambda_{v_0} \neq 1 \quad \text{at one bad place } v_0 \text{ only}, \qquad \lambda_v = 1 \quad (v \neq v_0)$$

be realized? If not, is the obstruction only the product formula, label synchronization, rigidity of a decomposition-group preserving global model, or another global constraint? We ask for the distinction between a purely global anabelian reading, using arithmetic fundamental-group data and decomposition groups, and a synchronized reading in which label synchronization of the type in [IUTchI, Prop. 6.5] is part of the given data.

If a one-place shear is realizable but compensated at other places so that the total is preserved, the form of compensation should be made explicit. Formally, a common ratio $r \neq 1$ at all places does not cancel in a nonzero total; what remains open is whether place-dependent families, and one-place support in particular, occur in the image of global differences.

Relation to (9-1): requires separate examination. (9-1) states membership for standard global objects; it does not classify realizability of prescribed local families. Under a fixed dictionary and Bridge (a), total consequences for the standard difference may follow, but not the one-place support condition.

Bearing on the central question. Tells whether a freedom in the degree survives global synchronization. Since such a freedom acts on all labels alike, this bears on how large the indeterminacies are and on the role of global data stressed in Step (xii) of the proof of Corollary 3.12 [IUTchIII, p. 185], rather than on a supplier directly.

Answer (author's reading). By the answer to Q0, a degree-moving action occurs only after realification, by a positive scalar. For the global realified Frobenioids of F_{mod}, compatibility with localization is stated up to $\mathbb{R}_{>0}$-orbits [IUTchII, Cor. 4.10(v)], that is, up to one scalar for all places. A family with $\lambda_{v_0} \neq 1$ and $\lambda_v = 1$ for $v \neq v_0$ would not preserve the product formula: it sends a principal divisor with a nonzero component at v_0 to a divisor of nonzero total degree. So one-place support is not realizable, and the obstruction is the product formula. Such a scalar acts on all labels alike.

Q2. Is the degree readout invariant under the standard difference of sections?

For the algebraic and theta sections in [EtTh], if the identification of coefficients is transported to the theta-section side, is the valuation-quotient readout of the q-parameter Kummer class invariant? Precisely, does $\varepsilon_v = D_v^{\mathrm{tr}}(q_v) - D_v(q_v)$ vanish? Is this already a direct consequence of Propositions 2.12 and 2.14(ii), Theorem 1.6(ii)(iii), and Corollary 2.19(i), or is an additional claim needed at the coefficient-identification step? Abstractly, invariance is immediate once the readout factors through a quotient on which the two splittings agree. The remaining issue is the source-level identification showing that the actual degree readout and transport have that form.

Relation to (9-1): connected under an additional identification. With the assembly dictionary and readout identification fixed, a numerical shadow follows from (9-1); the exact transformation rule inside [EtTh] does not. This question may close from existing [EtTh] results alone.

Bearing on the central question. With the dictionary and the readout identification fixed, (9-1) implies a numerical shadow of $\varepsilon_v = 0$ (see the relation above), so a nonzero value would close this route. The value 0 is consistent with the route, but it does not supply the labelwise switch.

Answer (author's reading). $\varepsilon_v = 0$, and it follows from the statements cited in the question. The difference of the two splittings is taken on the subquotient over $l \cdot \Delta_\Theta$ and yields the cyclotomic isomorphism [EtTh, Cor. 2.19(i)]; the automorphism obtained by shifting by a $\boldsymbol{\mu}_N$-valued difference of sections is the identity on both the quotient and the kernel [EtTh, Prop. 2.14(ii)]; and the valuation readout of Kummer classes of elements of the base field, such as q_v, is preserved by tempered isomorphisms together with $1 \in \widehat{\mathbb{Z}}$ [EtTh, Thm. 1.6(ii)]. Transporting the identification of coefficients to the theta-section side therefore leaves the readout of q_v unchanged, and no further claim is needed at the coefficient-identification step. This agrees with the abstract invariance checked in Lean (Section 2.5).

Q3. What is the smallest category containing both sections and the degree readout?

What is the smallest category in which the algebraic section, the theta section and the degree readout simultaneously exist as data while discrete rigidity is retained? In a mono-theta environment the algebraic section is only detected in subquotients through commutators, whereas a bi-theta environment has both sections but loses discrete rigidity. Is there a standard category resolving this tension, or evidence that the tension itself is essential?

Relation to (9-1): requires separate examination. Even if (9-1) is formulated in the LANA framework, that does not decide which mono/bi-theta environment is the minimal home for this question. It is a question at the level of source definitions.

Bearing on the central question. Locates the smallest category in which this route, and a supplier along it, could be stated.

Answer (author's reading). The mono-theta environment itself. The algebraic section is not part of its data, but the commutator property of [EtTh, Prop. 2.12] takes its place, and the three rigidities of [EtTh, Cor. 2.19] hold; [EtTh, Rem. 2.19.2] notes that the commutator property plays the role of the algebraic section, which does not appear in a mono-theta environment. Adding the algebraic section as data, as in a bi-theta environment, cuts the symmetry acting on the labels down to multiples of N [EtTh, Prop. 2.14(iii)]. In this sense the tension is essential, and [EtTh] resolves it by using the algebraic section as a property rather than as data. The degree readout is available through [EtTh, Thm. 1.6(ii)].

Q4. Does a global δ exist without Hodge theatres?

Can the projective system of local mono-theta environments and its η-classes be glued, from global arithmetic fundamental-group data and restrictions to decomposition groups alone, into one global object δ whose local components are not independent? If so, what functor or reconstruction algorithm gives the gluing? If not, what is the smallest additional datum required? Is the understanding correct that label synchronization alone does not give cocycle coherence?

Relation to (9-1): requires separate examination. A global object in the LANA framework would still need to be identified with a gluing of [EtTh] section systems. Whether the setup can be founded in pre-IUT vocabulary is not decided by granting (9-1).

Bearing on the central question. Decides whether the global datum that a supplier might use can be stated without Hodge theatres.

Answer (from the cited statements). Not in the naive sense. The source states that the local tempered fundamental groups at bad places do not extend to a global tempered fundamental group [IUTchIII, Cor. 3.12, proof, Step (viii), p. 180]. What is used instead is a global $\pm$-synchronization, which produces a profinite conjugacy indeterminacy in the vertically coric construction of the theta

values at torsion points; this indeterminacy is resolved by the theory of [IUTchI, §2] ([IUTchI, Rem. 6.12.4(iii)]; [IUTchII, Rems. 4.5.3(iii), 4.11.2(iii)]). So the additional datum is this synchronization, which is part of the Hodge-theatre apparatus. We found no statement that makes the local η-classes coherent as cocycles; the synchronization is at the level of conjugacy.

Q5. Can the degree action of δ be computed in a Tate prototype?

For a Tate curve over a finite p-adic field containing the necessary roots of unity, for example at $\ell = 5$ and including Gaussian-type local test cases, make explicit from theta series and product expansions the classes $\theta_{v,j}$ and the change of η-classes under translation by $a \in \mathbb{Z}$. Under the standard splitting obtained by restriction to the zero section, compute the transported degree action ε_v, separating classically known parts from parts depending on the choice of section.

Relation to (9-1): partly known; the centre requires separate examination. Valuations of theta values and the functional equation follow from classical computation and [EtTh, Props. 1.4–1.5]. The value of ε_v under the standard splitting is not computed by (9-1).

Bearing on the central question. A computation that tests Q2 in a Tate prototype.

Answer (author's reading). The transported degree action is $\varepsilon_v = 0$, by the answer to Q2. The profile itself comes from the quadratic exponents of the theta series and its functional equation $\ddot{\Theta}(q_X^{a/2}\ddot{U}) = (-1)^a q_X^{-a^2/2}\ddot{U}^{-2a}\ddot{\Theta}(\ddot{U})$ [EtTh, Prop. 1.4(ii)], through the fact that restricting the étale theta class to a point gives the value of the theta function there up to units [EtTh, Prop. 1.4(iii)]; this gives the valuations $\frac{j^2}{2\ell}\,\mathrm{ord}_v(q_v)$ of Appendix A.1. The difference of sections does not contribute to the profile. We did not tabulate the values at $\ell = 5$, since the answer does not depend on them.

Q6. Bridge (a): is the assembly two-path difference the [EtTh] section difference?

At each bad place, restrict the composite "Route 1 theta-link value part $\circ$ (Route 2 permutation transport)$^{-1}$" to the data of the mono-theta splitting monoid of [EtTh, Thm. 2.2(ii)]. Does it agree with transport induced by the [EtTh] section difference, the η-cocycle? Is that agreement compatible, including degree readouts, with the passage from the single-basepoint method to the procession method in [IUTchII, Rem. 3.5.3] and with horizontal compatibility in [IUTchIII, Cor. 2.3(iv)]? If this is immediate from a numbered statement, we would be grateful for that reference.

Relation to (9-1): corresponding under a fixed dictionary. With a dictionary between LANA objects and the assembly two paths, a positive answer largely overlaps the content of (9-1). But fixing that dictionary is itself an unproved bridge, and identification with the [EtTh] δ is not immediate from the statement of (9-1).

Bearing on the central question. A positive answer would reduce the central question, along this route, to properties of the [EtTh] section difference; a negative answer would close the route.

Answer (author's reading; it depends on Q2). In the strict sense, no. The two-path difference sends the labelwise readouts $\{j^2 u_v\}_j$ to $\{u_v\}_j$; that is the profile switch. The transport induced by the [EtTh] section difference leaves the degree readout of the q-datum unchanged (Q2). The two can agree on degree readouts only if all labels have the same readout, which fails for $\ell \geq 5$. So this route does not carry the switch, and the compatibility with [IUTchII, Rem. 3.5.3] and [IUTchIII, Cor. 2.3(iv)] does not arise for it. In the sense of inclusion in the hull, agreement up to the indeterminacies is (S) itself.

Q7. Bridge (b): what is needed to pass from equal readouts to object membership?

Under a splitting into unit part and a rational ray generated by a normalized element, can two pilot objects with the same unit part and the same total procession degree be treated as the same possible image in the hull-compatible sense? If not, what is the smallest additional datum beyond total degree: the profile tuple itself, a lower-bound or inclusion condition for regions, or another object-level identification?

Relation to (9-1): membership is supplied; the implication requires separate examination. Granting (9-1) supplies the needed membership itself. It does not prove an implication "equal readouts $\Rightarrow$ membership". A total degree is a many-to-one shadow of a tuple, and without additional structure such an implication fails in general. In the assembly-shaped countermodel, this insufficiency is visible when the transported number is not connected to a theta-side object.

Bearing on the central question. This is where the profile enters. The distinction between equal readouts and membership is now part of (S) in Section 1.3: a supplier of the value form need not supply the carrier form.

Answer (from the cited statements and Section 2.5). No. Equal unit parts and equal total procession degree do not make two pilot objects the same possible image: a total is a many-to-one shadow of a profile (Section 2.5; the model of [U26d]). For the value form of (S) an inequality of totals is enough; for membership, that is, the carrier form of (S) and (9-1) in the reading of [U26, Def. 4], an inclusion of regions is needed. That inclusion is the smallest additional datum, and it is what the central question asks for.

D Statement P′

Statement P′ is a revised presentation of an earlier reformulation, arranged to match the local skeleton and the decomposition into questions above. It is not claimed to be equivalent to the residual. P′ is not an additional independent problem beside Q0–Q7: Q0–Q5 separate its prerequisites, definitions and local core, while Q6–Q7 are bridges outside P′ leading back to the assembly.

Part of P′	Questions	Remark
Two sections and δ setup	Q3	Minimal category
Replacement by degree readouts	Q0, Q2, Q5	Shear classification, transported readout, computation
Globalization	Q1, Q4	Image of global differences; existence of global δ
Total compatibility assertion	Q1 standard case + Q2	Standard difference, not arbitrary shear families
Return to assembly	Q6, Q7	Bridges (a)(b), outside P′

D.1 Differences from a solved form

[EtTh, Cor. 2.19(i)] says that the difference of two sections gives a compatible isomorphism on cyclotomic objects. P′ differs in two ways: first, it replaces the object of compatibility by value-group degree readouts; second, it states the local theory globally, under one global δ and the product-formula constraint. The geometric Szpiro inequality in the function-field case is only an expository anchor for the second difference: there globality appears as a closed intersection-theoretic total. It does not provide a mechanism in the number-field case. For a prior formulation of a geometric Corollary 3.12, see [Jos].

D.2 Setup and statement

Take K, E/K, S, q_v, $\ell \geq 5$, theta torsion data and $\eta_{v,j}$ as in Appendix A. Take the two sections s^{alg}, s^{Θ} and their difference δ in the sense of Appendix A; cyclotomic compatibility is known by [EtTh, Cor. 2.19(i)]. As a marked **premise-or-part-of-the-question**, suppose that the sections and δ exist as one global object whose local sections are restrictions to decomposition groups. This globalization is not founded here from the local [EtTh] theory; it is Q4.

Statement 1 (P′: compatibility of transported degree readouts)**.** δ is compatible with value-group degree readouts. Precisely, for the system over all $v \in S$, the degree datum $\mathrm{ord}_v(q_v)$ read through the algebraic section agrees, as the same readout in the same reconstructed value group, with the degree datum of the same object read through the theta section, that is, from the system of Kummer classes $\{\eta_{v,j}\}$.

What is read through the theta section is not the total degree of the theta system itself, which is computed from its own valuations and does not equal the q-degree. It is the degree of q-data transported by δ. Nor is the assertion about capsulewise profiles $\{j^2\}$ versus $\{1\}$, or a place-by-place assertion. It is asserted only for the system over S, for one global δ, under the product-formula constraint.

Relation to (9-1): connected under additional identifications. With dictionary and Bridges (a)(b) fixed, the numerical content of P′ is obtained as a shadow of (9-1). The converse does not hold: P′ is a readout assertion, not object membership itself.

D.3 Rating of P′

No equivalence between P′ and (α) has been proved; the gap consists of Q6 and Q7. The globalization in the setup is an unverified seam, and a version with the global clause removed is not P′. The local skeleton gives formal scaffolding, but transplantation to degree readouts is itself the open content and does not follow automatically from Corollary 2.19(i), whose proof uses cyclotomic Kummer structure. The direction checked formally is (α) plus decomposition identities implying (β); numerical compatibility P′ alone does not imply (α). If P′ is false, this assembly route closes, but that would not disprove ABC. If P′ is true, Bridges (a)(b) and the IUT black box still remain. With the provisional answers of Appendix C, the local part of P′ holds (Q2), but Bridge (a) fails in the strict sense (Q6), so P′ does not lead to (α) along this route.

E Candidate forms A–D and the implication map

Before this question list, the residual (α) was examined in the following candidate forms. We record only their relation to the questions here. The implications are sketches under a fixed dictionary, not proved equivalences between autonomous statements.

Form	Content	Relation here
A	Commutativity, in the value component, of the two-path square	Assembly-side form of (α); Q6 asks for [EtTh]-side identification
B	Application of semantic principle S to the relevant isomorphism and value component	Semantic clothing of A; the general principle is not the question here
C	Commutativity of mono-anabelian reconstruction R with theta transport	Algorithmic clothing of A; seam: assembly degree data are output of R
D = (9-1)	Membership form of $\eta^q = \eta_S^{\mathrm{anab}}$ type	Current reference form; this note asks before and around it
P′	Global compatibility of transported degree readouts	Common-vocabulary entrance; weaker readout form than A

Under supplied inputs, the sketch is: the relevant instance of B corresponds to A and to (α); C would give A through its seam; and D corresponds to (α) under the fixed dictionary. These dictionary-dependent relations were not formalized in Lean. Q0–Q5 separate prerequisites and definitions outside this map; Q6–Q7 ask about its seams.

F Summary of relations with (9-1)

The labels below mean the position of a question if (9-1) is granted. They do not include the truth of (9-1) or correctness of a dictionary. This is a division-of-roles summary. Many questions retain independent meaning after (9-1), as category classification or foundations for globalization.

Question	Relation	Reason in brief
Central question (§1)	Supplier of (9-1)	A supplier of the carrier form of (S) gives (9-1) in the reading of [U26, Def. 4]; (9-1) gives the value form
Q0 local-shear classification	Separate	(9-1) gives no categorywise automorphism classification
Q1 globalization / one place	Separate	Standard membership does not classify prescribed families
Q2 transported-readout invariance	Additional identification	Numerical shadow follows; internal [EtTh] rule is separate
Q3 minimal category	Separate	Source-definition question
Q4 global δ	Separate	Identification with [EtTh] gluing is additional
Q5 Tate computation	Partial	Theta valuations known; ε_v is not computed by (9-1)
Q6 Bridge (a)	Corresponding under dictionary	Dictionary itself unproved
Q7 Bridge (b)	Membership supplied; implication separate	(9-1) does not prove readout-to-membership
P′	Additional identifications	Shadow under dictionary and bridges; converse fails
Forms A, D	Corresponding under dictionary	Rephrasings of (α)
Form B, general principle	Separate	One instance does not imply a general semantic principle
Form C	Additional identification	Assembly degree data as output of R is the seam

G Other places in the source

Besides the constructions named in the central question, the following places may bear on the edge of Section 1.5. The list is not exhaustive.

Place	Content	Bearing
[IUTchIII, Thm. 3.11(i)], pp. 153–155	Multiradial representation; at the end of (i), the permutation symmetries of the étale-like structures	How the output is made independent of the holomorphic structure
[IUTchIII, Thm. 3.11(ii)], pp. 155–156	Log-Kummer correspondence; (Ind3)	Step (xi-d) calls the column-one log-Kummer correspondence essential
[IUTchIII, Thm. 3.11(iii)], pp. 156–158	Compatibility of the Kummer isomorphisms with the $\Theta^{\times\boldsymbol{\mu}}_{\mathrm{LGP}}$-links	Relates the Θ-link to Kummer theory
[IUTchIII, Rem. 3.11.1(iii)–(v)], pp. 160–163	(IPL), (SHE), (APT) and the hidden internal structure	The properties that Step (xi) keeps, and what it sets aside
[IUTchIII, Rem. 3.11.1(vi)–(viii)], pp. 163–167	Naive models: distinct labels give (IPL) but leave (SHE) unclear; confusing q with its powers gives a trivial multiradial representation; the features of Theorem 3.11 on which the estimate depends; the permutation symmetry $\Pi \to G \leftarrow \Pi$	The source's own statement of the dilemma, and of how the theory is meant to avoid it
[IUTchIII, Rem. 3.9.5(vii)–(x)], pp. 131–145	Hull, (Ob1)–(Ob9), (sQ4), the closed loop of (viii)(ix)	The passage from regions to comparable numbers, cited in (xi-c)–(xi-e)
[IUTchIII, Cor. 3.12, proof], Steps (i)–(x), pp. 175–181, and (xii), p. 185	Set-up of the comparison; in (xii), the role of the global realified Frobenioids in the $\Theta^{\times\boldsymbol{\mu}}_{\mathrm{LGP}}$-link	Step (xii) stresses that global data matter in Step (xi)
[IUTchIII, Rem. 3.12.2], pp. 187–195	Interpretation of the gluing and of the entanglement of addition and multiplication	Context of (xi-a)
[IUTchII, Cor. 4.5(iv)(v)], p. 133; [IUTchII, Cor. 4.6(iv)(v)], pp. 138–139; [IUTchII, Cor. 4.10(v)], p. 160	Theta and Gaussian monoids and Frobenioids, the ratio vector $(j^2)_j$, compatibility up to $\mathbb{R}_{>0}$-orbits	The source of the profile
[EtTh, Thm. 1.6, Props. 2.12, 2.14, Cor. 2.19]	Mono-theta environments, the difference of the theta and algebraic sections, cyclotomic rigidity	The local skeleton of Appendix B; part of the structure set aside in (xi-b)
[FrdI, Thm. 6.4]	Restoration of the degree scale in tiers: $\mathbb{R}_{>0}$, $\mathbb{Q}_{>0}$, 1	What an equivalence of realified Frobenioids fixes: a single scalar on one line
[Form, §§3–4]	Symmetry of the étale Hodge theatre under exchanging the domain and the codomain of the Θ-link with the basic prime-strip fixed; the four triangles; (APT) in the second and third	The author's own decomposition of the step

Disclosure

This is personal work by an independent researcher, unrelated to any institutional affiliation. Generative AI was used under the author's direction for auditing, formalization and drafting. Responsibility for the contents rests with the author. This is a work report not yet independently re-verified.

References

[IUTchI] S. Mochizuki, Inter-universal Teichmüller Theory I: Construction of Hodge Theaters, *Publ. Res. Inst. Math. Sci.* 57 (2021), 3–207.

[IUTchII] S. Mochizuki, Inter-universal Teichmüller Theory II: Hodge-Arakelov-theoretic Evaluation, *Publ. Res. Inst. Math. Sci.* 57 (2021), 209–401.

[IUTchIII] S. Mochizuki, Inter-universal Teichmüller Theory III: Canonical Splittings of the Log-theta-lattice, *Publ. Res. Inst. Math. Sci.* 57 (2021), 403–626.

[IUTchIV] S. Mochizuki, Inter-universal Teichmüller Theory IV: Log-volume Computations and Set-theoretic Foundations, *Publ. Res. Inst. Math. Sci.* 57 (2021), 627–723.

[EtTh] S. Mochizuki, The Étale Theta Function and its Frobenioid-theoretic Manifestations, *Publ. Res. Inst. Math. Sci.* 45 (2009), 227–349.

[FrdI] S. Mochizuki, The Geometry of Frobenioids I: The General Theory, *Kyushu J. Math.* 62 (2008), 293–400.

[FrdII] S. Mochizuki, The Geometry of Frobenioids II: Poly-Frobenioids, *Kyushu J. Math.* 63 (2009), 1–182.

[AbsTopIII] S. Mochizuki, Topics in Absolute Anabelian Geometry III: Global Reconstruction Algorithms, *J. Math. Sci. Univ. Tokyo* 22 (2015), 939–1156.

[AbsAnab] S. Mochizuki, The Absolute Anabelian Geometry of Hyperbolic Curves, in *Galois Groups and Fundamental Groups*, MSRI Publ. 41, Cambridge Univ. Press, 2003, 77–122.

[SS] P. Scholze and J. Stix, Why abc is still a conjecture, 2018.

[NSW] J. Neukirch, A. Schmidt, and K. Wingberg, *Cohomology of Number Fields*, 2nd ed., Springer, 2008, Thm. 7.5.14.

[HN] Y. Hoshi and Y. Nishio, On the outer automorphism groups of the absolute Galois groups of mixed-characteristic local fields, *Res. Number Theory* 8 (2022), no. 3, Paper No. 56.

[Kon] K. Kondo, Anabelian aspects of the outer automorphism groups of the absolute Galois groups of mixed-characteristic local fields, arXiv:2512.09231v2.

[Jos] K. Joshi, Construction of Arithmetic Teichmuller Spaces $\mathrm{II}_{1/2}$: Deformations of Number Fields, arXiv:2305.10398.

[JosII] K. Joshi, Construction of Arithmetic Teichmuller Spaces II: Proof of a local prototype of Mochizuki's Corollary 3.12, arXiv:2303.01662v3, Thms. 6.9.1 and 9.2.1. Not treated here as evaluated input.

[Zen26] C. Zeng, An Integral-Structure Inflation in the Hull Operation of IUT Corollary 3.12, Hexagon 2610.00646v1. Not verified or used as input here.

[LANA] Project LANA, *Project LANA interim report on IUT theory*, July 2026, 50 pp. Mentioned as a reference form; its results are not used here as proved inputs.

[lana] Project LANA, *lana-agents/iut*, Lean repository, `https://github.com/lana-agents/iut`; files `Iut/WallEquiv/NonDerivable.lean` (added 13 October 2026, commit 180c049) and `Iut/Wall/Constraints.lean`, and the EtaleTheta, Wall and LogLink modules, consulted at commit 553076a (3 November 2026). Mentioned as nearby formalization; its results are not used here as proved inputs.

[Higuchi] J. R. Higuchi, A formal negative result for the passage from Step (xi-e) to Step (xi-f) in IUT III, preprint, March 31, 2026, 9 pp., DOI 10.13140/RG.2.2.10989.76006.

[Pajo] P. Pajo, Scale-free common-container comparisons do not entail numerical inequalities: a minimum-viable no-go theorem motivated by the IUT III Theorem 3.11-to-Corollary 3.12 bridge, preliminary research note, April 26, 2026, 14 pp., DOI 10.13140/RG.2.2.33620.16005.

[Takkun] Takkun-kohinata, *IUT_LEAN*, Lean repository, `https://github.com/Takkun-kohinata/IUT_LEAN`, commit 9d56c46 (15 July 2026); files `IutLean/Multiradial/ThetaHullDichotomy.lean` and `DisputedCore.lean`.

[Saito] Y. Saito, A registered-provenance audit of IUT III, Corollary 3.12, Step (xi), version 2.0.0, Zenodo, 26 July 2026, DOI 10.5281/zenodo.21596198.

[sfingali] sfingali, The Scholze–Stix objection, formally: the arithmetic is agreed, the q-side is absent, Zenodo, 2 August 2026, DOI 10.5281/zenodo.21757640.

[Geiger] L. Geiger, The bridge certificate: a line-level audit instrument for inter-theoretic transfer arguments, Zenodo, first version 14 August 2026, DOI 10.5281/zenodo.21925803 (all versions).

[Form] S. Mochizuki, On the formalization of IUT: a preliminary progress report (joint work in progress with Y. Hoshi, G. Yamashita, Y. Yang), slides, April 2026, 14 pp.

[Ri] D. Richards, *motizukipdf — Mochizuki's papers, plus a toolkit for the data types of IUT*, GitHub repository `https://github.com/oneman/motizukipdf`; Lean port `lean/IUT/LeanForm.lean` at commit 2665db1 (30 October 2026).

[DHPS] T. Dupuy and A. Hilado, Probabilistic Szpiro, baby Szpiro, and explicit Szpiro from Mochizuki's Corollary 3.12, arXiv:2004.13108v2 [math.NT], 29 April 2020.

[U26] T. Urahigashi, Where Theorem 3.11 sits, Zenodo, DOI 10.5281/zenodo.24104932 (v1.2.1).

[U26b] T. Urahigashi, Forgetting and Completeness, Zenodo, DOI 10.5281/zenodo.24210290 (v1.1.5).

[U26c] T. Urahigashi, A conditional assembly from Theorem 3.11 to Corollary 3.12: decomposition of the scalar layer, a semantic principle, and isolation of the value-group profile switch, Zenodo, DOI 10.5281/zenodo.24209900 (v1.0.3).

[U26d] T. Urahigashi, Scale preservation is not enough: a profile-level independence countermodel for the Corollary 3.12 assembly inputs, Zenodo, DOI 10.5281/zenodo.24209904 (v1.0.1).

[SL] T. Urahigashi, Segment or loop? A homotopy-type-theoretic overview of the dispute over Corollary 3.12 of inter-universal Teichmüller theory, version 1.4.0, Zenodo, DOI 10.5281/zenodo.24039360.